

Computing Derivatives for Logistic Regression

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Motivation

- Objective: compute derivatives needed for gradient descent in logistic regression.
- Model for one example: $z = W^T X + b$, $A = \sigma(z) = \frac{1}{1+e^{-z}}$.
- Loss (binary cross-entropy):
$$L(A, Y) = -[Y \log A + (1 - Y) \log(1 - A)].$$

Forward Propagation

$$z = W^T X + b,$$

$$A = \sigma(z) = \frac{1}{1 + e^{-z}},$$

$$L(A, Y) = -\left(Y \log A + (1 - Y) \log(1 - A)\right).$$

Backward: Derivative w.r.t. A

Starting from the loss: $L(A, Y) = -[Y \log A + (1 - Y) \log(1 - A)]$

$$\frac{\partial L}{\partial A} = -\left(\frac{Y}{A} - \frac{1 - Y}{1 - A}\right) = -\frac{Y}{A} + \frac{1 - Y}{1 - A}.$$

Backward: Derivative w.r.t. z

Using the chain rule and $\frac{\partial A}{\partial z} = A(1 - A)$:

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial A} \cdot \frac{\partial A}{\partial z} = \left(-\frac{Y}{A} + \frac{1 - Y}{1 - A} \right) A(1 - A) = A - Y. \quad ()$$

Key result: the error term is simply $\partial L / \partial z = A - Y$.

Gradients w.r.t. Parameters (Single Example)

Let $X \in \mathbb{R}^n$, $W \in \mathbb{R}^n$, scalar b .

$$\begin{aligned}\frac{\partial L}{\partial W} &= X (A - Y), \\ \frac{\partial L}{\partial b} &= A - Y.\end{aligned}$$

Component-wise for features X_i : $\frac{\partial L}{\partial W_i} = X_i (A - Y)$.

Mini-batch / Dataset Averaging

For a batch of size m with examples $(X^{(i)}, Y^{(i)})$:

$$z^{(i)} = W^T X^{(i)} + b, \quad A^{(i)} = \sigma(z^{(i)}),$$

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(A^{(i)}, Y^{(i)}),$$

$$\frac{\partial J}{\partial W} = \frac{1}{m} \sum_{i=1}^m X^{(i)} (A^{(i)} - Y^{(i)}),$$

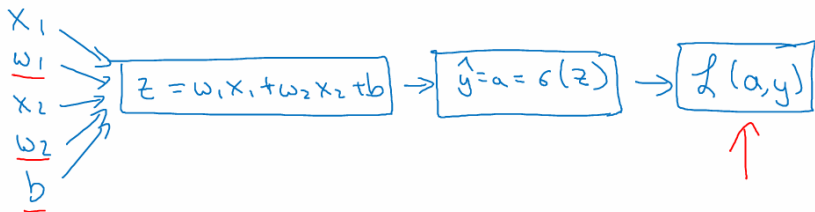
$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (A^{(i)} - Y^{(i)}).$$

Logistic regression recap

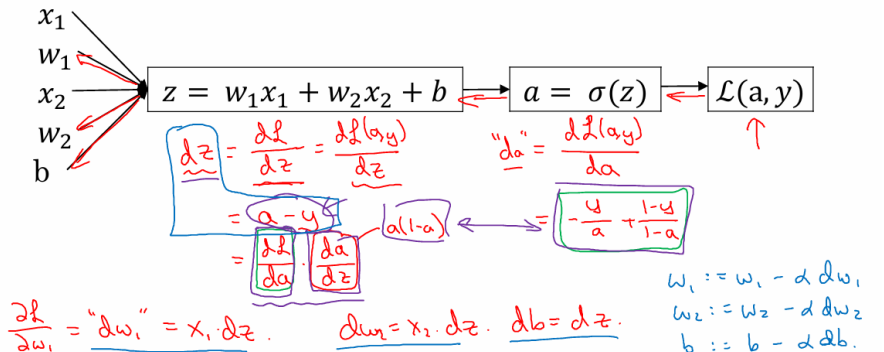
$$\rightarrow z = w^T x + b$$

$$\rightarrow \hat{y} = a = \sigma(\underline{z})$$

$$\rightarrow \mathcal{L}(a, y) = -(y \log(a) + (1 - y) \log(1 - a))$$



Logistic regression derivatives



Andrew Ng

Gradient Descent Updates

With learning rate $\alpha > 0$:

$$\begin{aligned}W &\leftarrow W - \alpha \frac{\partial J}{\partial W}, \\b &\leftarrow b - \alpha \frac{\partial J}{\partial b}.\end{aligned}$$

Interpretation: move parameters opposite to the gradient to reduce the cost.

Key Takeaways

Quantity	Formula	Meaning
$\frac{\partial L}{\partial z}$	$A - Y$	Error at the logit z
$\frac{\partial L}{\partial W}$	$X(A - Y)$	Weight gradient (per feature)
$\frac{\partial L}{\partial b}$	$A - Y$	Bias gradient
Update	$\theta \leftarrow \theta - \alpha \nabla_{\theta} J$	Descent step